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THE COMPOSITION OF THE INTESTINAL MICROBIOTA AFTER ALLOGENEIC HAEMATOPOIETIC STEM CELL TRANSPLANTATION AND ITS ASSOCIATION WITH GRAFT VERSUS HOST DISEASE AS ASSESSED BY 16SRIBOSOMAL RIBONUCLEIC ACID

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To observe the evolution of the intestinal microbiota in patients after allogeneic hematopoietic stem cell transplantation (allo-HSCT) and discuss the relationship between the intestinal microbiota and graft-versus-host disease (GVHD). In this study, 11 patients who underwent allo-HSCT in the Aerospace Central Hospital from January 2021 to October 2021 were selected, along with 11 donors. Fecal specimens were collected 7 times: at admission, after pre-treatment, and every 3 weeks after transplantation from patients and once from donors. The composition of the intestinal microbiota and its association with GVHD after allogeneic hematopoietic stem cell transplantation were analyzed by 16S rRNA sequencing. Of the 11 patients, 5 developed GVHD, and 6 did not. The diversity of the intestinal microbiota among GVHD patients first increased and then decreased after transplantation, while that among non-GVHD patients first increased and then tended to be stable. The diversity of the intestinal microbiota among GVHD patients was lower than that among non-GVHD patients before pre-treatment and after transplantation. The taxa diversity of the intestinal microbiota in the non-GVHD group was better than that in the GVHD group before allo-HSCT, and the difference was statistically significant ($P < 0.05$ for OTUs and CHAO1 index). The taxa abundance of *Enterococcaceae* 2.16% (2.13%, 2.22%) before allo-HSCT was significantly higher than that in the non-GVHD group 1.33% (0.27%, 1.52%), and the difference was statistically significant ($P = 0.004$). There was no significant difference between the GVHD group and the non-GVHD group in the diversity of the intestinal microbiota of donors ($P > 0.05$). The characteristics of the intestinal microbiota in the final sample of patients in the GVHD group were similar to the preoperative structure of the intestinal microbiota. In conclusion: The decrease in the diversity of the intestinal microbiota after HSCT may be a risk factor for the occurrence of GVHD. The presence of *Enterococcaceae* in the intestinal microbiota may be associated with an increased risk of developing GVHD. The intestinal microbiota reconstitute to be close to the intestinal microbiota composition of the donors in the non-GVHD group.

Key words: *allogeneic hematopoietic stem cell transplantation, intestinal microbiota, graft-versus-host disease, taxonomic bacterial diversity, ribosomal ribonucleic acid, immune system*

INTRODUCTION

Hematopoietic stem cell transplantation (HSCT) is a radical treatment for multiple hematological diseases and has been carried out widely. Graft-versus-host disease (GVHD), one of the main complications of allogeneic HSCT, has high morbidity and mortality and has a severe impact on the therapeutic effect of HSCT. Among all allogeneic hematopoietic cell transplant patients, 30–50% develop aGVHD (grade I–IV) (1, 2). GVHD is the second largest cause of mortality after HSCT (22%) after the recurrence of the primary disease (29%) (3). The intestine, rich in immune cells, is considered the largest immune organ, making it one of the main organs involved in GVHD - intestinal GVHD is present

in almost all deaths caused by GVHD (4). Growing evidence suggests the critical role of the intestinal microbiota in immune regulation. GVHD is closely related to intestinal microbiota disturbance. Studies have revealed that the composition of the intestinal microbiota in patients after HSCT undergoes significant changes, including a decrease in overall diversity, overgrowth of certain dominant bacteria, such as *Enterococcus* and *Clostridium*, and a decrease in beneficial bacteria such as *Lactobacillus* and *Blautia* (5). The incidence of GVHD increased among patients with a low diversity of fecal microbiota, with increased GVHD-related mortality. However, some studies have inconsistent results. A study found that oral vancomycin prophylaxis, which is very disruptive to the microbiome, effectively prevented CDI in recipients who had

undergone allo-HSCT but was not associated with an increased risk of GVHD (6) and data on the intestinal microbiota of donors are lacking. In this study, we used 16S rRNA sequencing to detect the intestinal microbiota of donors and patients. In addition, the alterations in intestinal microbiota diversity and abundance following allogeneic hematopoietic stem cell transplantation are discussed, the impact of intestinal microbiota on GVHD after allo-HSCT is revealed, and the difference in intestinal microbiota between donors and patients after allo-HSCT is explored.

MATERIALS AND METHODS

Participants

The study was conducted in accordance with the Declaration of Helsinki (as was revised in 2013) and was approved by the Ethics Committee of the Aerospace Central Hospital. Written informed consent was obtained from all participants. (20191228).

Eleven patients, including 8 males and 3 females, with a median age of 29 years (10–55 years) who underwent allo-HSCT in the Aerospace Central Hospital from January 2021 to October 2021 were selected, along with 11 donors, for this study. Inclusion criteria were: individuals willing to provide written informed consent and participants who were otherwise well were included in the study. They underwent their first transplant, and the exact contact with the donor was made. Only patients on a modified Bu/Cy conditioning regimen were included.

Data acquisition

The occurrence and severity of GVHD after allo-HSCT was recorded. GVHD diagnosis was made based on the extent of involvement and symptoms of target organs (skin, intestine, lung, and liver) and the Seattle BMT Center grading standards (7).

Specimen collection

Fecal specimens were collected 7 times: at admission, after pre-treatment, and every 3 weeks after transplantation from patients and from donors before mobilization of stem cells. The specimens were stored at -80° within 2 h after collection.

Microbial diversity analysis

Microbial DNA was extracted from fecal samples using the Qiagen DNA extraction Kit 51604 (Germantown, MD, USA) according to the manufacturer's protocols. The V3-V4 region of the bacteria 16S ribosomal RNA genes were amplified by PCR (95°C for 3 min, followed by 30 cycles at 98°C for 20 s, 58°C for 15 s, and 72°C for 20 s and a final extension at 72°C for 5 min) using primers 341F 5'-CCTACGGGRCAGCAG-3' and 806R 5'-GGACTACVGGGTATCTAATC-3'. PCR reactions were performed in a 30 μ L mixture containing 15 μ L of 2 $\times$ KAPA Library Amplification ReadyMix, 1 μ L of each primer (10 μ M), 50 ng of template DNA and ddH₂O. For this study, we performed paired-end sequencing using Illumina technology on the HiSeq PE250 platform. Amplicons were extracted from 2% agarose gels and purified using the AxyPrep DNA Gel Extraction

Table 1. Clinical characteristics of patients.

Patients	Diagnosis	Age	Gender	Relationship with donor	Age of donor	Combined disease	GVHD
A	Acute lymphoblastic leukemia	16	Male	Father	38	Ewing's sarcoma	+30 lung
B	Acute myeloblastic leukemia	45	Male	Elder brother	48	Chronic hepatitis C carrier	+64 intestine
C	T/myeloid mixed leukemia	10	Male	Father	33	Infective fever	+14 intestine, +20 skin
D	Myelodysplastic syndrome	35	Male	Younger male cousin	30	Intestinal infection, pulmonary infection	+103 intestine, skin
E	Acute lymphoblastic leukemia	29	Female	Younger brother	25	Pulmonary infection	+82 liver
F	Acute lymphoblastic leukemia	25	Male	Father	46	Thrombocytopenia at the age of four	None
G	Myelodysplastic syndrome (MDS-RAEB)	51	Male	Son	24	Pulmonary tuberculosis, hepatic tuberculosis	None
H	Acute lymphoblastic leukemia	52	Female	Younger brother	54	Cholecystitis, pneumonia	None
I	Acute myelogenous leukemia	39	Female	Son	14	Intestinal infection, pulmonary infection	None
J	T-lymphoblastic lymphoma	27	Male	Father	51	Pulmonary infection	None
K	Acute myelogenous leukemia	14	Male	Father	40	Pulmonary infection, pulmonary tuberculosis	None

Kit (Axygen Biosciences, Union City, CA, USA) according to the manufacturer's instructions and quantified using Qubit®2.0 (Invitrogen, Waltham, MA, USA). After preparation of the library, these tags were sequenced on the HiSeq platform (Illumina Inc., Hayword, CA, USA) for paired-end reads of 300/250 bp, which overlapped on their 3 ends for concatenation into original longer tags. DNA extraction, library construction and sequencing were conducted at Realbio Genomics Institute (Shanghai, China). Tags, trimmed of barcodes and primers, were further checked on their rest lengths and average base quality. 16S tags were restricted between 220 bp and 500 bp such that the average Phred score of bases was no worse than 20 (Q20) and no more than 3 ambiguous N. The copy number of tags was enumerated and redundancy of repeated tags was removed. Only the tags with frequencies greater than 1, which tended to be more reliable, were clustered into OTUs, each of which had a representative tag. Operational Taxonomic Units were clustered with 97% similarity using UPARSE (<http://drive5.com/uparse/>), and chimeric sequences were identified and removed using Userach (version 7.0). Each representative tag was assigned to a taxon by RDP Classifier (<http://rdp.cme.msu.edu/>) against the RDP database (<http://rdp.cme.msu.edu/>) using a confidence threshold of 0.8. The OTU profiling table and alpha/beta diversity analyses were also achieved by python scripts of QIIME.

Statistical analysis

We used SPSS 21.0 statistical software to conduct descriptive analysis of the data. To test the parameters of the diversity of the intestinal microbiota, we performed the Wilcoxon signed-rank test and Mann-Whitney U test. We considered $P < 0.05$ to be statistically significant. Specifically, we used the CHAO1 index to measure the abundance and evenness of the microbiota and evaluated the species richness of the intestinal microbiota using the operational taxonomic unit (OTU) mean.

RESULTS

Clinical characteristics of patients

Eleven patients were included in this study. Among them, 5 patients developed GVHD (GVHD group, $n=5$), and 6 did not

develop GVHD (non-GVHD group, $n=6$), as shown in Table 1. Five patients with grade II–III GVHD received intravenous methylprednisolone at an initial dose of 1 mg/kg/d for approximately 1 week, and then the dose was reduced by 10–20 mg/d weekly until it was stopped. All 11 patients were treated with broad-spectrum antibiotics, including Imipenem-Cilastatin or Meropenem, after allo-HSCT. Broad-spectrum antibiotics are often used to prevent infection because the body's immune system is destroyed after pre-treatment. We enrolled patients who had been treated with Imipenem-Cilastatin or Meropenem due to fever. The baseline characteristics of demographics and pre-HSCT parameters between the GVHD and on-GVHD groups were shown in Table 2.

The characteristics of the intestinal microbiota during the HSCT transplantation process

The species richness of the intestinal microbiota was evaluated by the operational taxonomic unit (OTU) mean, as shown in Fig. 1. The CHAO1 index, a taxa alpha diversity index, was used to measure the abundance and evenness of the microbiota. The Fig. 2 displays the temporal changes in the Chao1 index for each patient across multiple timepoints, including pre-transplantation, pre-processing, and 3–15 weeks after transplantation. We conducted further analyses of the GVHD and non-GVHD groups and found statistically significant differences in the Chao1 index at 9 weeks ($P=0.030$) and 15 weeks ($P=0.004$) post-HSCT as is shown in Fig. 3. The taxa diversity of the microbiota in the initial sample of GVHD patients was low; it reached the lowest level after pre-treatment and then increased; it then decreased again and was lower than the level in the patients before transplantation; the taxa diversity of the intestinal microbiota in the final sample of GVHD patients was significantly lower than that of donors. The taxa diversity of the intestinal microbiota in the initial sample of non-GVHD patients was high; it reached the lowest level after pre-treatment and then gradually increased until it tended to be stable, but it was lower than the level in the patients before transplantation and lower than the level of donors. In the final sample of GVHD patients, at the family level, the taxa abundance of *Lachnospiraceae* 3.00% (0.99%, 4.09%) significantly decreased compared with that before transplantation 21.67% (17.54%, 22.89%), and the difference was statistically significant ($P=0.043$). The taxa abundance of *Lactobacillaceae* showed no

Table 2. Comparisons of baseline characteristics of patients.

Variables	non-GVHD, (n=5) Median (IQR)	GVHD, (n=6) Median (IQR)	P-value
Age	33.0 (22.3, 51.3)	29.0 (13.0, 40.0)	0.537
Male, n (%)	4 (66.7)	4 (80.0)	1.000
OTUs	318.00 (173.00, 419.00)	96.00 (95.50, 111.00)	0.004
Chao1	152.57 (64.00, 430.76)	130.38 (127.28, 145.19)	0.429
PD_whole_tree	18.78 (9.35, 29.62)	9.83 (9.59, 9.85)	0.082
Shannon	4.9300 (0.6735, 5.4143)	3.3304 (3.3244, 3.8958)	0.537
Simpson	0.8988 (0.1633, 0.8993)	0.8492 (0.8483, 0.8830)	0.931
Goods_coverage	0.9990 (0.9987, 0.9992)	0.9987 (0.9986, 0.9989)	0.662
<i>Enterobacteriaceae</i> (%)	49.52 (0.77, 94.54)	44.86 (33.34, 49.26)	0.004
<i>Bacteroidaceae</i> (%)	28.65 (15.80, 96.25)	0.74 (0.74, 11.06)	0.030
<i>Lachnospiraceae</i> (%)	5.19 (4.89, 5.23)	21.67 (20.88, 22.70)	0.004
<i>Enterococcaceae</i> (%)	1.33 (0.28, 1.52)	2.14 (2.13, 2.16)	0.004
<i>Lactobacillaceae</i> (%)	2.00 (1.00, 2.00)	0.00 (0.00, 0.00)	0.024

significant changes compared with that before transplantation, and both were extremely low. The taxa abundance of *Fusobacteriaceae* 0.004% (0.001%, 0.008%) significantly decreased compared with that before transplantation 1.01% (1.00%, 1.59%), and the difference was statistically significant ($P=0.043$). In the final sample of non-GVHD patients, the taxa abundance of *Bacteroidaceae* 36.27% (28.22%, 54.41%) showed little change compared with that before transplantation 28.65% (14.59%, 96.27%); the taxa abundance of *Lachnospiraceae* 18.67% (14.78%, 24.81%) significantly

increased compared with that before transplantation 5.19% (3.86%, 5.24%), and the difference was statistically significant ($P=0.028$); the taxa abundance of *Lactobacillaceae* 5.05% (4.00%, 5.74%) significantly increased compared with that before transplantation 2.00% (1.00%, 2.00%), and the difference was statistically significant ($P=0.028$); the taxa abundance of *Fusobacteriaceae* 5.10% (4.20%, 5.21%) significantly increased compared with that before transplantation 1.12% (1.05%, 1.66%) , and the difference was statistically significant ($P=0.028$), as shown in Table 3.

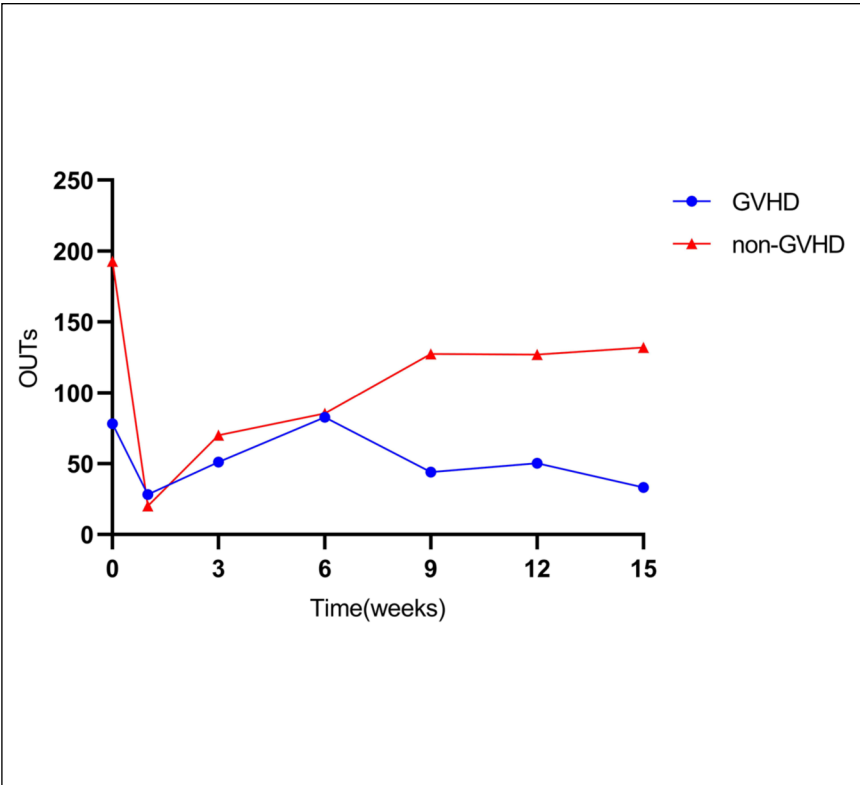


Fig. 1. Change trend of OTUs in the GVHD group and non-GVHD group. Point 0 on the timeline represents pre-HSCT; point 1 on the timeline represents conditioning. The initial sample of patients with graft-versus-host disease (GVHD) had a low count of operational taxonomic units (OTUs) in their microbiota. This count reached its lowest level after pre-treatment and then increased, but subsequently decreased again and remained lower than the level observed in patients before transplantation. On the other hand, the initial sample of patients without GVHD had a high count of OTUs in their intestinal microbiota. This count reached its lowest level after pre-treatment and then gradually increased until it tended to stabilize, although it remained lower than the level observed in patients before transplantation. Notably, starting from post-HSCT week 6, the non-GVHD group consistently displayed higher operational taxonomic unit (OTU) counts compared to the GVHD group. This suggests that the non-GVHD group had a higher diversity of microbiota, as indicated by the higher OTU counts, compared to the GVHD group from post-HSCT week 6 onward.

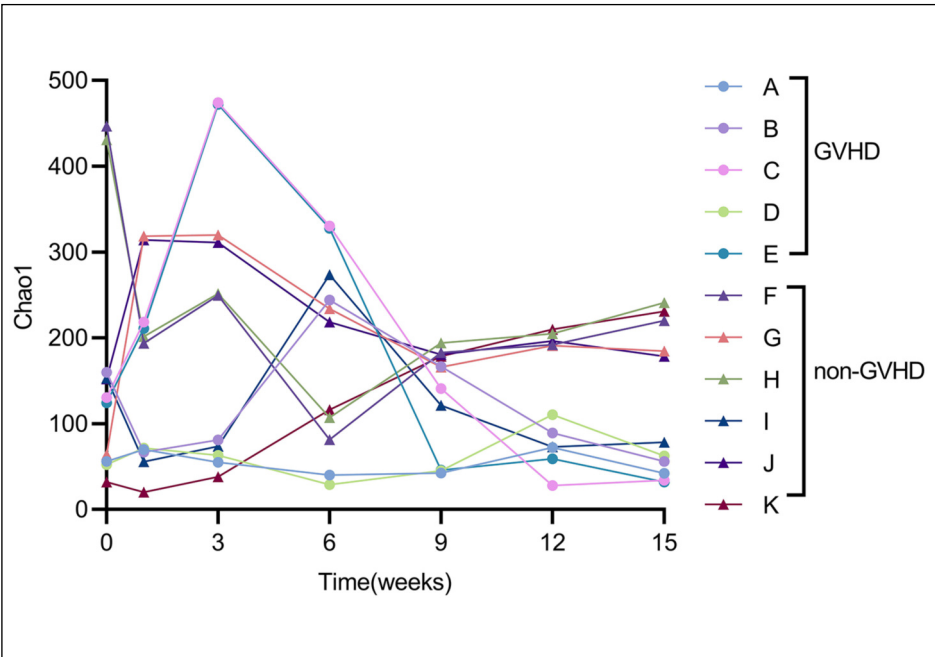


Fig. 2. Change trend of the Chao1 index in every patient. Point 0 on the timeline represents pre-HSCT; point 1 on the timeline represents conditioning. The temporal changes in the Chao1 index for each patient across multiple timepoints, including pre-transplantation, pre-processing, and 3–15 weeks after transplantation. The Chao1 index, which reflects microbial diversity, showed distinct patterns during the transplantation process. Overall, the non-GVHD group had higher Chao1 index values after transplantation compared to the GVHD group. In the GVHD group, some patients showed significant changes in their Chao1 indexes during the transplantation process, but ultimately tended to have lower values compared to the non-GVHD group.

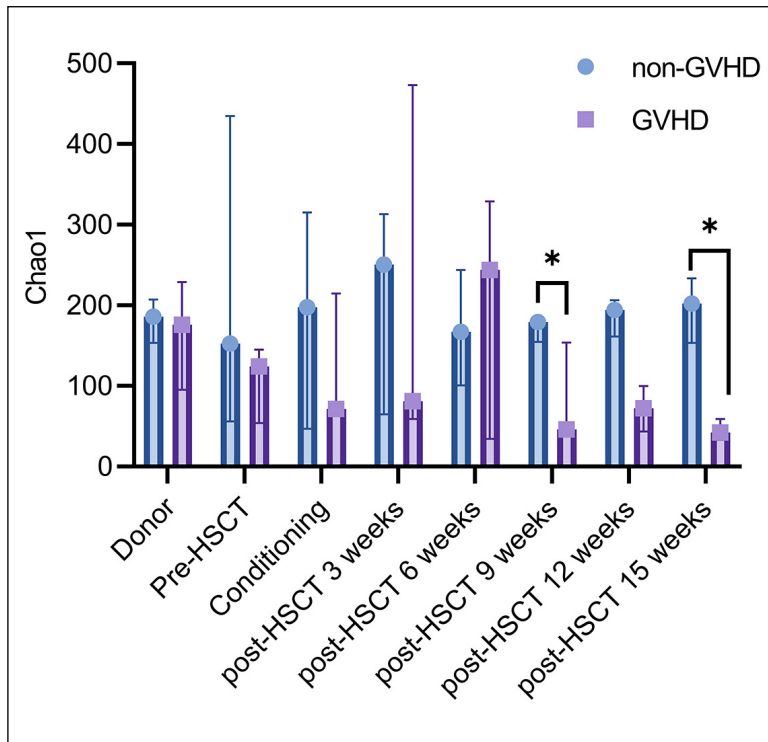


Fig. 3. Change trend of the CHAO1 index between GVHD and non-GVHD groups. * represents $P < 0.05$. The Chao1 index was compared between the GVHD group and the non-GVHD group throughout the HSCT process, including the Chao1 index values from the donors of both groups. The results of statistical analysis showed significant differences in the Chao1 index between the two groups at 9 weeks ($P = 0.030$) and 15 weeks ($P = 0.004$). This indicates that there were statistically significant variations in the microbial diversity, as reflected by the Chao1 index, between the GVHD and non-GVHD groups at these specific time points during the HSCT process.

Table 3. Comparison of relative species abundance of certain bacteria between the graft-versus-host disease (GVHD) and non-GVHD patients.

	GVHD (n=5) Median (IQR)			non-GVHD (n=6) Median (IQR)		
	Pre-HSCT	Post-HSCT week 15	P-value	Pre-HSCT	Post-HSCT week 15	P-value
<i>Lachnospiraceae</i> (%)	21.67 (17.54, 22.89)	3.00 (0.99, 4.09)	0.043	5.19 (3.86, 5.24)	18.67 (14.78, 24.81)	0.028
<i>Enterococcaceae</i> (%)	2.16 (2.13, 2.22)	2.18 (2.00, 2.87)	0.500	1.33 (0.27, 1.52)	0.011 (0.000, 0.014)	0.028
<i>Lactobacillaceae</i> (%)	0.00 (0.00, 0.00)	0.004 (0.000, 0.031)	1.000	2.00 (1.00, 2.00)	5.05 (4.00, 5.74)	0.028
<i>Fusobacteriaceae</i> (%)	1.01 (1.00, 1.59)	0.004 (0.001, 0.008)	0.043	1.12 (1.05, 1.66)	5.10 (4.20, 5.21)	0.028
<i>Bacteroidaceae</i> (%)	0.69 (0.02, 11.06)	28.19 (0.01, 35.05)	0.225	28.65 (14.59, 96.27)	36.27 (28.22, 54.41)	0.917

Table 4. Parameters of species diversity of patients and donors before and after transplantation.

	pre-HSCT			post-HSCT week 15			Donor		
	Median (IQR)		P-value	Median (IQR)		P-value	Median (IQR)		P-value
	nonGVHD	GVHD		nonGVHD	GVHD		nonGVHD	GVHD	
OTUs	318.00 (173.00, 419.00)	96.00 (95.50, 111.00)	0.004	167.00 (48.00, 192.00)	36.00 (30.50, 40.00)	0.082	192.00 (183.00, 197.00)	203.00 (193.00, 205.00)	0.662
Chao1	152.57 (64.00, 430.76)	130.38 (127.28, 145.19)	0.429	186.17 (154.62, 203.17)	229.10 (202.55, 229.10)	0.931	216.13 (154.62, 400.88)	229.10 (202.55, 229.10)	0.329
PD_whole_tree	18.78 (9.35, 29.62)	9.83 (9.59, 9.85)	0.082	14.45 (13.24, 15.10)	4.87 (4.67, 4.89)	0.004	11.53 (11.01, 11.65)	12.29 (11.87, 13.52)	0.931
Shannon	4.9300 (0.6735, 5.4143)	3.3304 (3.3244, 3.8958)	0.537	4.5969 (4.4910, 4.6883)	1.5691 (1.5682, 2.3076)	0.004	4.4698 (4.3544, 4.6586)	4.1194 (3.804, 4.1195)	0.030
Simpson	0.8988 (0.1633, 0.8993)	0.8492 (0.8483, 0.8830)	0.931	0.9246 (0.9152, 0.9327)	0.5836 (0.5836, 0.6775)	0.004	0.9246 (0.9152, 0.9246)	0.7162 (0.6499, 0.7867)	0.017
Goods_coverage	0.9990 (0.9987, 0.9992)	0.9987 (0.9986, 0.9989)	0.662	0.9985 (0.9983, 0.9992)	0.9996 (0.9996, 0.9996)	0.009	0.9990 (0.9988, 0.9991)	0.9987 (0.9987, 0.9988)	0.662

Comparison between the graft-versus-host disease group and non-graft-versus-host disease group before allo-HSCT

The taxonomic diversity of the intestinal microbiota in the non-GVHD group was better than that in the GVHD group before allo-HSCT, and the difference was statistically significant ($P < 0.05$ for OTUs), as shown in Table 4. In the GVHD group, the taxa abundance of *Bacteroidaceae* 0.74 (0.74, 11.06) was significantly lower than that in the non-

GVHD group 28.65 (15.80, 96.25), and the difference was statistically significant ($P = 0.030$); the taxa abundance of *Lactobacillaceae* 0.00 (0.00, 0.00) was significantly lower than that in the non-GVHD group 2.00 (1.00, 2.00), and the difference was statistically significant ($P = 0.024$); the taxa abundance of *Enterococcaceae* 2.14 (2.13, 2.16) before allo-HSCT was significantly higher than that in the non-GVHD group 1.33 (0.28, 1.52), and the difference was statistically significant ($P = 0.004$), as shown in Table 5.

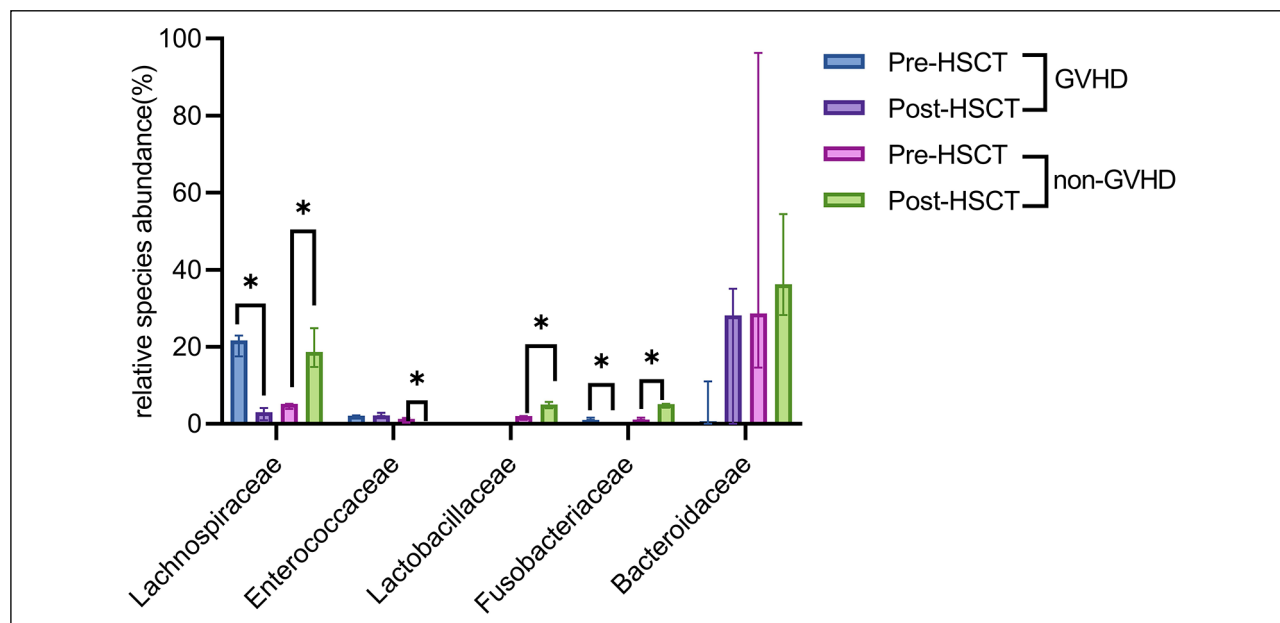


Fig. 4. Comparison of taxa abundance average before and after HSCT. * represents $P < 0.05$. At the family level, changes in taxa abundance were observed in different groups. In the GVHD group, the abundance of *Lachnospiraceae* and *Fusobacteriaceae* significantly decreased after transplantation compared to before transplantation, with a statistically significant difference ($P < 0.05$). The abundance of *Lactobacillaceae* showed no significant changes and remained low. In the non-GVHD group, the abundance of *Bacteroidaceae* showed little change after transplantation compared to before transplantation. However, the abundance of *Lachnospiraceae*, *Lactobacillaceae*, and *Fusobacteriaceae* significantly increased after transplantation compared to before transplantation, with a statistically significant difference ($P < 0.05$).

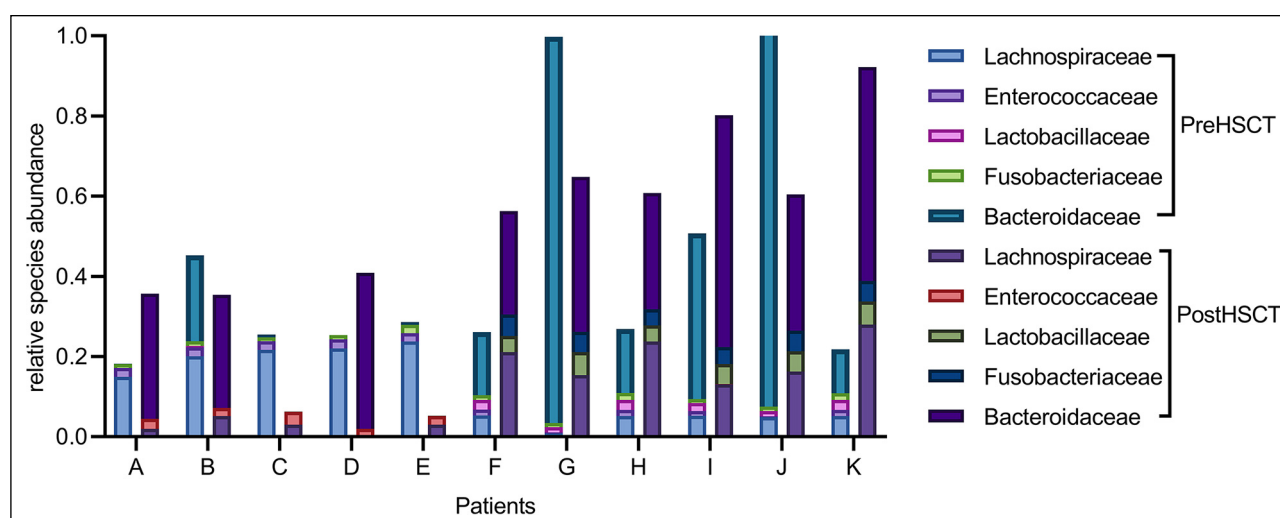


Fig. 5. Comparison of the individual microbiota composition. This figure visually presents the relative species abundance of intestinal microbiota before and after transplantation for each individual. The purpose of this figure is to observe the characteristics of changes in relative species abundance before and after transplantation for each individual. Statistical analysis for the changes in species abundance before and after transplantation is provided in Fig. 4 and Table 3. No statistical analysis is presented in this figure.

Table 5. Comparison of relative taxa abundance of certain bacteria between patients and donors before and after transplantation.

	pre-HSCT Median (IQR)			post-HSCT week 15 Median (IQR)			Donor Median (IQR)		
	nonGVHD	GVHD	P-value	nonGVHD	GVHD	P-value	nonGVHD	GVHD	P-value
<i>Enterobacteriaceae</i> (%)	49.52 (0.77, 94.5)	44.86 (33.34, 49.26)	0.004	3.31 (1.93, 38.9)	43.29 (36.26, 49.30)	0.052	0.08 (0.08, 0.10)	0.11 (0.03, 16.35)	1.000
<i>Bacteroidaceae</i> (%)	28.65 (15.80, 96.25)	0.74 (0.74, 11.06)	0.030	36.27 (29.04, 53.27)	0.008 (0.008, 14.101)	0.177	16.77 (15.50, 50.10)	52.59 (32.99, 52.59)	0.082
<i>Lachnospiraceae</i> (%)	5.19 (4.89, 5.23)	21.67 (20.88, 22.70)	0.004	18.67 (15.34, 23.75)	3.00 (3.00, 4.09)	0.004	17.11 (11.33, 23.04)	8.22 (8.22, 8.61)	0.030
<i>Enterococcaceae</i> (%)	1.33 (0.28, 1.52)	2.14 (2.13, 2.16)	0.004	0.01 (0.00, 0.01)	2.18 (2.10, 2.70)	0.004	0.00 (0.00, 0.01)	0.00 (0.00, 0.00)	0.126
<i>Lactobacillaceae</i> (%)	2.00 (1.00, 2.00)	0.00 (0.00, 0.00)	0.024	5.05 (4.00, 5.73)	0.03 (0.02, 0.03)	0.004	0.00 (0.00, 0.00)	0.00 (0.00, 0.00)	0.662

Table 6. Comparison of relative species abundance of *Enterococcaceae* after pre-treatment between the GVHD group and non-GVHD group.

	GVHD	Non-GVHD	P-value
<i>Enterococcaceae</i> (%)	57.48±37.77	11.31±1.70	0.004

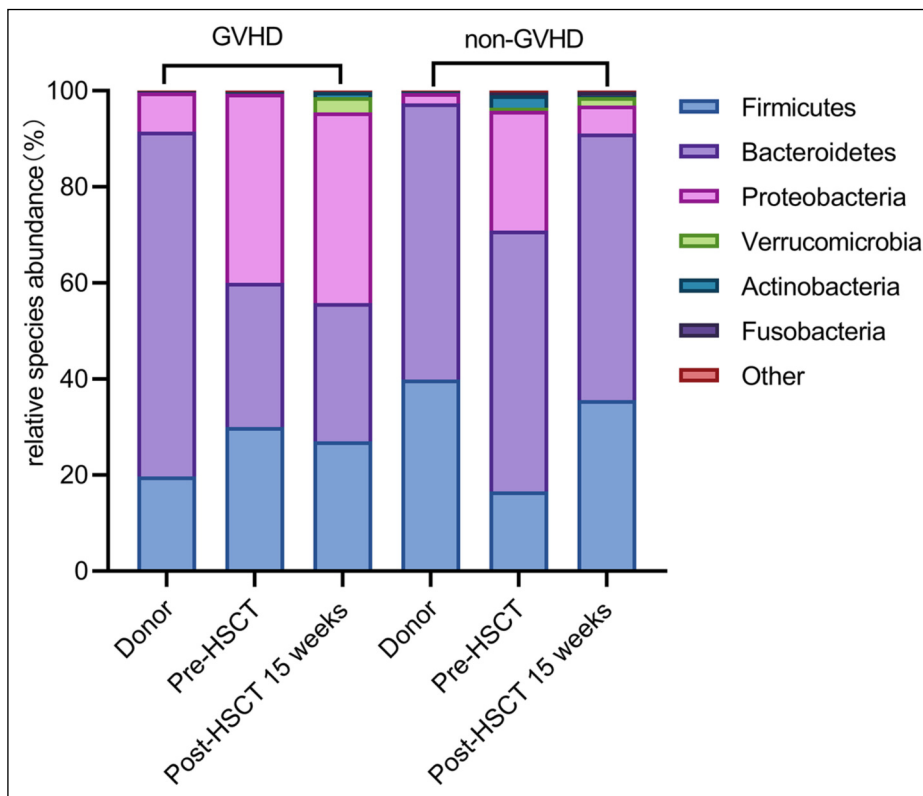


Fig. 6. Comparison of the taxa abundance of the intestinal microbiota at the phylum level. In the final sample of GVHD patients, the composition ratio of intestinal microbiota at the phylum level was similar to that of the patients before transplantation. However, in the final sample of non-GVHD patients, the composition ratio of intestinal microbiota at the phylum level was similar to that of the donors.

Comparison between the graft-versus-host disease group and non-graft-versus-host disease group after transplantation

The taxa diversity of intestinal microbiota in the non-GVHD group was better than that in the GVHD group after transplantation, and there were significant differences in PD_whole_tree, Shannon index, and Simpson index (all $P < 0.05$), as shown in Table 4. In the GVHD group, at the family level, the taxa abundance of *Lachnospiraceae* 3.00 (3.00, 4.09)

was significantly lower than that in the non-GVHD group 18.67 (15.34, 23.75), and the difference was statistically significant ($P = 0.004$); the taxa abundance of *Enterococcaceae* 0.01 (0.00, 0.01) was significantly higher than that in the non-GVHD group 2.18 (2.10, 2.70), and the difference was statistically significant ($P = 0.004$); the taxa abundance of *Lactobacillaceae* 0.03 (0.02, 0.03) was significantly lower than that in the non-GVHD group 5.05 (4.00, 5.73), and the difference was statistically significant ($P = 0.004$), as shown in Table 5. After pre-treatment, the OTU

and CHAO1 index means in the GVHD group were higher than those in the non-GVHD group, but the difference was not statistically significant. After pre-treatment (conditioning), which refers to the chemotherapy and/or radiation therapy that is given to a patient before the actual transplantation of stem cells, at the family level, the taxa abundance of *Enterococcaceae* ($57.48 \pm 37.77\%$) in the GVHD group was significantly higher than that in the non-GVHD group ($11.31 \pm 1.70\%$), and the difference was statistically significant ($P=0.004$), as shown in Table 6 and Figs. 4, 5.

Comparison of intestinal microbiota characteristics between two groups of donors in hematopoietic stem cell transplantation

There was no significant difference in the taxa abundances of *Enterobacteriaceae*, *Bacteroidaceae*, *Enterococcaceae*, and *Lactobacillaceae* (all $P>0.05$) between the two groups, as shown in Table 5. In the final sample of GVHD patients, the composition ratio of intestinal microbiota at the phylum level was similar to that of the patients before transplantation, as shown in Fig. 6; in the final sample of non-GVHD patients, the composition ratio of intestinal microbiota at the phylum level was similar to that of donors, as shown in Fig. 6.

DISCUSSION

Allo-HSCT is most likely to cure recurrent or refractory leukemia (8). It is also risky. Pre-treatment may cause damage to organs, and allogeneic reactions may cause GVHD. GVHD is the second largest cause of mortality. There is a close interaction between the intestinal microbiota and the immune system, while transplantation-related complications are closely associated with the immune system of the host. Therefore, the intestinal microbiota may influence the prognosis of allo-HSCT recipients. The intestinal microbiota is severely damaged and shows a gradual recovery trend with the restoration of the immune system during HSCT. Studies have indicated that the diversity of the microbiota tends to recover after allo-HSCT (9). However, there are still significant differences in the structure and composition of the microbiota compared with those before treatment (10). Enterococcus-related diarrhea may occur after allo-HSCT (11).

As early as the 1970s, Van Bekkum *et al.* and Jones *et al.* proposed the impact of intestinal microbiota on GVHD, finding that mice growing in a sterile environment were less likely to develop intestinal GVHD (12, 13). The analysis of intestinal microbiota characteristics in hematopoietic stem cell transplantation (HSCT) is critical in understanding the role of microbiota in the development of graft-versus-host disease (GVHD). In this study, we observed a significant shift in the microbiota community over time in both the GVHD and non-GVHD groups. The overall trend in the GVHD group showed that the operational taxonomic unit (OTU) was lowest at pre-treatment, increased at post-HSCT week 3, peaked at post-HSCT week 6, and tended to decrease at post-HSCT week 9. On the other hand, the non-GVHD group showed that the OTU was the lowest at conditioning, then gradually increased, peaked at post-HSCT week 9, and then tended to be stable.

Our findings suggest that the intestinal microbiota undergoes dynamic changes during HSCT and that these changes may be associated with the development of GVHD. The observed decrease in OTU abundance in the GVHD group at post-HSCT week 9 may suggest a possible link between microbiota dysbiosis and GVHD. Interestingly, the non-GVHD group showed an increase in OTU abundance at post-HSCT week 9, which may suggest a role for certain microbiota strains in promoting graft-host tolerance. Further studies are needed to elucidate the mechanisms underlying

the observed changes in microbiota community and to identify specific bacterial strains that may contribute to GVHD or promote graft-host tolerance. However, our study provides valuable insights into the temporal dynamics of intestinal microbiota during HSCT and highlights the need for more research in this area to improve clinical outcomes for HSCT patients. In an effort to decontaminate the intestinal microbiota prior to the study, a regimen of 0.5 grams of Levofloxacin administered daily for a period of one week was used. However, our analysis of the gut microbiota in the GVHD group showed that the number of operational taxonomic units (OTUs) remained high after pre-treatment, suggesting that intestinal decontamination might not have been fully effective. This could potentially contribute to an increased risk for GVHD, as alterations in the gut microbiota have been shown to play a role in the pathogenesis of this condition. Previous studies have reported that complete intestinal decontamination may prevent the occurrence of GVHD (12, 14). Some studies have suggested that allo-HSCT recipients with significantly damaged intestinal microbiota are more likely to develop severe GVHD (15-18). The conclusions are inconsistent. Shono *et al.* proposed that drug-resistant bacteria, including drug-resistant *Enterococcus*, made it difficult to complete intestinal decontamination (19). In this study, we compared the relative taxa abundances of *Enterococcus* in the GVHD group and non-GVHD group after pre-treatment, and found that the abundance was significantly higher in the GVHD group than in the non-GVHD group, which supported the opinion of Shono (19). The incidence of GVHD significantly decreased in patients with successful intestinal decontamination (20). The results of our study indicate that in the GVHD group, the taxonomic diversity of the intestinal microbiota was high at pre-treatment, and in the non-GVHD group, the taxonomic diversity was the lowest at pretreatment. Based on the results of our research, it can be concluded that conditioning treatment prior to stem cell transplantation can lead to changes in the composition of the gut microbiota. Specifically, the abundance of *Enterococcaceae* at the family level was significantly higher in the GVHD group compared to the non-GVHD group. This suggests that *Enterococcaceae* may play a role in the development of GVHD following stem cell transplantation. Further research is needed to investigate the potential mechanisms underlying this association and to determine if interventions aimed at modulating the gut microbiota could be used to prevent or treat GVHD. Additionally, it would be useful to explore the potential clinical implications of these findings, such as the use of targeted antibiotics or probiotics to modulate the gut microbiota. The results of an experimental animal study (21) indicated that the intestinal microbiota of mice with GVHD shifted to *Escherichia coli*, *Enterococcus*, *Bacteroides*, and *Prevotella*; similar changes were also observed in some studies on enteritis (22-24) and patients with inflammatory bowel disease (25, 26). It is worth noting that the increase in *Escherichia coli* load was significantly correlated with the mortality of GVHD and the pathological injury of the intestine (17). In this study, we found that the *Fusobacteriaceae* content increased in the non-GVHD group and decreased in the GVHD group after transplantation and that the content in the GVHD group was significantly lower than that in the non-GVHD group after transplantation. Furthermore, recent research has revealed that conditions such as TNBS colitis can lead to a reduction in the abundance of bacterial taxa that produce short-chain fatty acids (SCFAs), including *Ruminococcaceae* and *Lachnospiraceae*. However, promisingly, studies have shown that the administration of a synbiotic therapy called Syngut can counteract this effect. In rats with TNBS colitis combined with cold stress, Syngut has been found to ameliorate gut inflammation. Notably, this beneficial effect of Syngut was accompanied by an increase in anti-inflammatory taxa, specifically *Ruminococcaceae* and *Lachnospiraceae* (27). This suggests that modulating the gut

microbiota with Syngut may hold potential for alleviating stress-related intestinal inflammation, particularly in patients with inflammatory bowel disease (IBD) who may experience intestinal disorders in conjunction with stress. These findings support the idea that an increase in *Lachnospiraceae* content in the intestinal microbiota may be associated with a potential reduction in the risk for GVHD, as *Lachnospiraceae* is known to be a SCFA producing bacterial family with anti-inflammatory effects. Therefore, the production of SCFAs by bacterial taxa such as *Ruminococcaceae* and *Lachnospiraceae* may play a beneficial role in improving the nutritional state of the intestinal mucosa, supporting the proliferation of intestinal epithelial cells, and potentially alleviating inflammatory reactions, thereby reducing the risk for GVHD and stress-related intestinal inflammation in patients with IBD.

The results of our study showed that the *Lachnospiraceae* content increased in the non-GVHD group and decreased in the GVHD group after transplantation, suggesting that the increase in *Lachnospiraceae* content in the intestinal microbiota was associated with a potential reduction in the risk for GVHD (*Blautia* belongs to *Lachnospiraceae*). Jenq *et al.* (28) found that a decrease in the *Lactobacillus* content in mice before bone marrow transplantation may lead to GVHD, and the reintroduction of *Lactobacillus* may alleviate GVHD. The results of our study showed that the *Lactobacillaceae* content in the GVHD group was lower than that in the non-GVHD group before and after transplantation, and the *Lactobacillaceae* content in the non-GVHD group tended to increase after transplantation. The content of *Lactobacillaceae* in the intestinal microbiota was observed to increase in the non-GVHD group; however, our study did not investigate the direct association of *Lactobacillaceae* with the reduction in the risk of GVHD development. After pre-treatment (conditioning), the taxa abundance of *Enterococcus* in the GVHD group was significantly higher than that in the non-GVHD group, suggesting that the presence of *Enterococcus* in the intestinal microbiota may be associated with an increased risk of GVHD. This is consistent with the research of Stein-Thoeringer *et al.* (29). However, further studies are needed to determine whether reintroducing *Lactobacillaceae* can alleviate GVHD.

The immune system and intestinal microbiota were damaged during allo-HSCT, and a new immune system was constructed after allo-HSCT. The new immune system was derived from the donor. The immune system could be selective to the intestinal microbiota, while the intestinal microbiota could regulate the immune system. In addition, the impact of autologous blood transfusion on bone marrow hematopoiesis was investigated in the animal experiment, comparing the effects of autologous whole blood transfusion and autologous blood component transfusion, such as PABD (30). The results showed that autologous blood component transfusion had less inhibitory effect on bone marrow hematopoiesis compared to autologous whole blood transfusion, suggesting that PABD or similar blood component transfusion may be more suitable for transplantation in the animal model. It is speculated that the immune system of the donor may have excellent immune tolerance to the original intestinal microbiota of the recipient, thus reducing the occurrence of GVHD. Kakiha *et al.* (31) suggested that fecal transplantation was safe and reliable for allo-HSCT recipients and was effective for GVHD. The results of an experimental animal study (32) indicated that changing the intestinal microbiota of the donor had no impact on the occurrence of GVHD, and the intestinal microbiota of the donor did not affect the induction and severity of GVHD. Our study found no significant difference in the intestinal microbiota of donors between the GVHD and non-GVHD groups. Additionally, the characteristics of the intestinal microbiota in the non-GVHD group were similar to those of donors after transplantation.

While we hypothesize that the immune system constructed after transplantation may have a strong immune tolerance and a mild inflammatory reaction to the microbiota, further research is needed to confirm this potential mechanism for avoiding GVHD. This was an observational study without the intervention of the clinical treatment process; therefore, a further study could be conducted on whether the occurrence of GVHD can be reduced in HSCT recipients who receive fecal transplants from the same individual.

We concluded that the reduced diversity of the intestinal microbiota after HSCT may be a risk factor for GVHD. The presence of *Enterococcaceae* in the intestinal microbiota may be associated with an increased risk of developing GVHD. Our findings revealed phylum-level reconstitution of the intestinal microbiota in both the GVHD and non-GVHD groups post-transplantation, with a stronger resemblance to the donor's microbiota composition observed in the non-GVHD group. These results suggest a possible correlation between microbiota reconstitution and a lower incidence of GVHD. The *Fusobacteriaceae*, *Lachnospiraceae*, and *Lactobacillaceae* content increased, which may reduce the risk for GVHD.

This was an observational study in which clinical treatment was not intervened, the interference factors of intestinal microbiota could not be controlled, and there were many confounding factors. The sample size was small, and clinical studies with large sample sizes are required to verify the conclusion. Some speculations should be further discussed based on animal experiments.

Abbreviations: allo-HSCT, allogeneic haematopoietic stem cell transplantation; BMT, bone marrow transplantation; DNA, deoxyribonucleic acid; GVHD, graft versus host disease; OUT, operational taxonomic unit; PCR, polymerase chain reaction; PD, phylogenetic diversity; rRNA, ribosomal ribonucleic acid; SPSS, statistical product and service solutions; PD, phylogenetic diversity

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